

Hip and Thigh Contusions and Strains

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CONTUSIONS

Contusions are the most common injuries to the hip, thigh, and pelvis. Collisions with other athletes and falls to the ground are the most common causes of contusions. Contusions can be superficial and limited to the subcutaneous tissue, or they can be deep, involving the bone, muscle, and ligaments. When they are associated with muscle involvement, contusions can result in slow bleeding with significant hematoma formation. Depending on the depth of the contusion and the extent of the injury, symptoms may occur immediately or after a 24- to 48-hour delay. Occasionally the force of the impact can cause a shearing of the connection between the muscle and subcutaneous tissue. Often, particularly with superficial contusions, treatment is of short duration and is based on the patient's symptoms. Patients generally have a rapid return to play (RTP). With the diagnosis of a deep contusion, however, the treatment period may be extended. Aside from icing, treatment should be delayed for 48 hours to ensure that all bleeding has stopped. After 48 hours, treatment with antiinflammatory drugs, heat, massage, and physical therapy is implemented. Rehabilitation should be aimed at maintaining flexibility and muscle mass. Range of motion (ROM) should be monitored because patients are prone to the development of myositis ossificans after sustaining deep contusions involving the muscle.

Iliac Crest Contusion

Iliac crest contusion, commonly known as a hip pointer, is an anterior pelvic contusion that often affects athletes involved in contact sports. These contusions result from either a direct fall onto the iliac crest or from a direct blow, as seen in football, hockey, and soccer (Fig. 87.1)^{1,2}

History

The athlete will almost immediately note pain over the iliac crest and/or greater trochanter after a fall or collision. He or she will note an inability to ambulate without a limp and pain with side-to-side or attempted crossover movements.³

Physical Examination

Examination of the injured athlete will reveal an antalgic gait, pain with palpation of the pelvic brim, bruising, and swelling. Active ROM of the hip will be decreased. Strength testing will

demonstrate a marked decrease in any or all of the following muscle groups, depending on the location of the contusion: the hip flexors, sartorius and rectus femoris, internal and external obliques, tensor fascia lata, gluteus medius, latissimus dorsi, and paraspinal muscles.^{3,4}

Imaging

Diagnosis of a hip pointer is made primarily on the basis of the history and physical examination. Radiographs are used only to rule out further injury, such as a fracture (Fig. 87.2). Rarely, magnetic resonance imaging (MRI) or ultrasound are needed; however, some investigators have advocated their role in the evaluation of hematoma formation in an effort to quantify RTP.³

Treatment Options

Nonoperative treatment with protection, rest, ice, compression, and elevation (PRICE) is central to the treatment of hip pointers. Crutches may be used for the first few days. Ice, rest, and compression should be instituted for the first 48 hours to decrease the risk of hematoma formation. Antiinflammatory agents may also be used during the first 5 to 7 days. Beginning on day 3, the athlete may begin painless ROM activities. Some investigators have advocated delivery of a cortisone injection to the affected area to decrease pain and swelling; however, the risks of such injections must be weighed. Physical therapy may be initiated once a painless ROM has been established. Therapy should include stretches, sport-specific massage, and strengthening. Surgery is rarely indicated for hip pointer injuries; however, concomitant injuries such as sports hernias must be ruled out.⁵



Authors' Preferred Technique

Iliac Crest Contusion

Our preferred technique is to initiate PRICE and use of crutches for the initial 24 h. A repeat physical examination in the office should be undertaken within 3–4 days of the injury. Provided that an avulsion fracture is not suspected, ROM exercises and gentle stretches are begun. A localized cortisone injection is considered for athletes who are not progressing as rapidly as anticipated. In those instances, activities are stopped for 48–72 h, after which rehabilitation is resumed in earnest. RTP is expected in 2–4 weeks once the athlete has completed strength testing and sport-specific drills.

Abstract

The muscles that surround the hip are vulnerable to injury from direct impact (contusion) or forceful contraction of a stretched muscle (strain). This chapter covers contusions and strains around the hip joint as well as special considerations for the pediatric athlete. Hamstring injuries are covered in Chapter 88.

Keywords

hip pointer
quadriceps contusion
rectus strain
adductor strain
ASIS avulsion
AIIS avulsion



Fig. 87.1 Clinical photograph of a hip pointer in a soccer athlete.

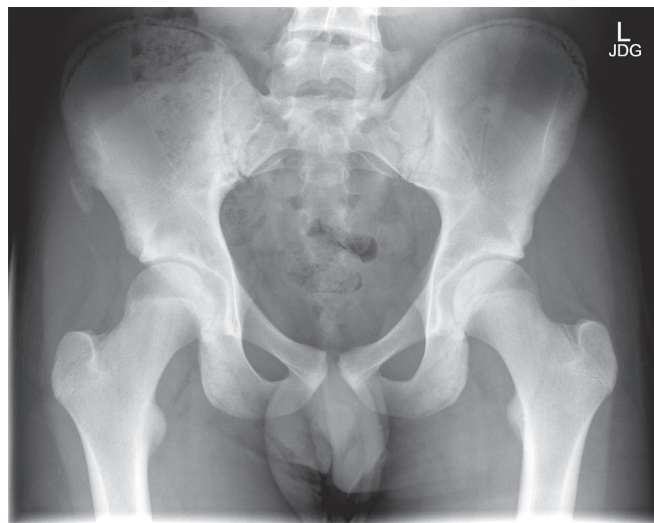


Fig. 87.2 Pelvic radiograph demonstrating an avulsion fracture of the anterosuperior iliac spine.

Return to Play

RTP is anticipated 2 to 4 weeks after the time of injury. Sport-specific drills and strength testing of the affected limb must be completed prior to resumption of competitive activities. Upon returning to activities, appropriate padding should be worn to prevent reinjury.^{3,6}

Results

The treatment of hip pointers has yielded excellent results. Almost all athletes are able to return to their previous level of competition and performance once the injury has sufficiently healed.^{3,4,7}

TABLE 87.1 Classification of Quadriceps Contusions

Grade	Range of Motion
Mild	>90 degrees
Moderate	45–90 degrees
Severe	<45 degrees

Complications

Complications after hip pointers are rare. Those most commonly seen include hematoma formation, cutaneous nerve injury, or myositis ossificans. Missed fractures or avulsions injuries are rare complications.^{3,5,7} Occasionally, as a result of severe impact, there can be an internal degloving injury, known as a Morel-Lavallée lesion, whereby the subcutaneous tissue is stripped of its connection to the underlying fascia. The resulting hematoma can be painful and may expand to fill a large cavity. Untreated, this cavity can form a cystic lesion, allowing the fluid collection to recur, and the hematoma can even serve as a nidus for bacterial colonization and infection. In the acute setting, conservative management—with cold therapy, compression, early ROM, and observation—is appropriate. If symptoms worsen or persist, advanced imaging studies can be helpful in directing treatment options, which include aspiration and either percutaneous or open drainage. A high index of suspicion is necessary to diagnose this condition, which is rare but has been reported in National Football League (NFL) players after a direct blow or slide on the ground.^{8–10}

Quadriceps Contusion

The quadriceps consists of the rectus femoris and the vastus musculature (medialis, lateralis, and intermedius). Quadriceps contusions are the result of a direct blow or trauma to the thigh. They typically occur in the anterior or lateral compartment and are most commonly associated with football, soccer, rugby, or high-speed traumas. The pathophysiology associated with quadriceps contusions involves microtrauma to the muscle fibers, with resultant swelling and edema. More significant contusions can also result in hemorrhage formation, myositis ossificans, and even tears.

Jackson and Feagin¹¹ established a classification system based on knee flexion. They describe the muscle contusion as mild, moderate, or severe. Forty-eight hours after injury, mild contusions are able to flex greater than 90 degrees; moderate contusions, from 45 to 90 degrees; and severe contusions, less than 45 degrees (Table 87.1). This classification helps guide counseling and treatment, and resolution occurs on days 6, 56, and 72, respectively. No corresponding MRI classification system has been developed. Persons with injuries classified as moderate to severe were significantly more likely to experience myositis ossificans than were those with mild contusions.^{11,12}

History

Typically the patient will present on the field with a chief complaint of pain, lack of movement, or explosiveness and an antalgic

gait. Twenty-four to 48 hours later, the patient may note similar or amplified symptoms, with noticeable limitations of knee ROM, thigh pain, loss of function, and swelling.¹² As previously stated, the mechanism of injury is usually collision with a helmet or a severe kick to the thigh.

Physical Examination

Physical examination begins by evaluating the gait of the patient as he or she walks. Typically an antalgic gait is noted in persons with a moderate to severe contusion. A localized physical examination will note a tender, swollen, and often ecchymotic thigh in the area corresponding to the blunt trauma that was sustained. The thigh should be compared with the contralateral side as well. Palpation of the affected thigh may yield a palpable defect in persons with severe contusions, indicating a partial or complete tear. In addition, a firm thigh compared with the contralateral side has been associated with longer recovery times.¹³ However, the most important component of the physical examination is the ROM of the knee, because this component correlates with the severity of the injury and the anticipated return to activity.^{12,13} Rarely, symptoms of an anterior thigh compartment syndrome can be identified on the basis of the physical examination.¹³

Imaging

Plain radiographs may be taken in the initial setting to rule out fracture; however, they typically have a low yield when a thorough history and physical examination is conducted. Radiographs, to evaluate for developing myositis ossificans, may have a more substantial role 2 to 4 weeks after injury if a firm mass is palpated.^{14,15}

MRI and ultrasound may also be used to evaluate the extent of the muscular edema and hemorrhage (Fig. 87.3); however,

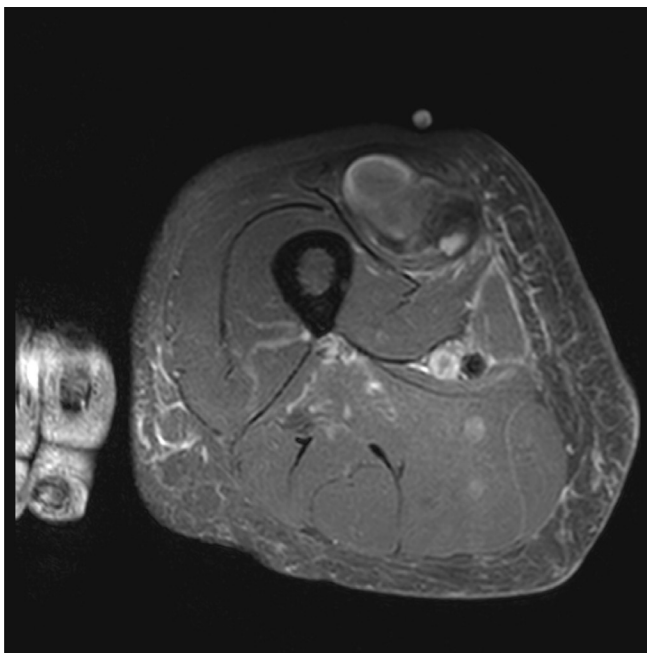


Fig. 87.3 Cross-sectional T2 magnetic resonance image of the thigh, demonstrating a developing hematoma in the body of the rectus femoris.

no level I studies have shown that imaging evaluations will accurately correlate with the severity of the contusion or predict RTP better than the Jackson and Feagin knee flexion classification.^{11,13,16,17}

Treatment Options

Nonoperative treatment. The treatment of quadriceps contusions is usually managed in a nonoperative or conservative fashion. Depending on the severity of the injury, some athletes may be able to continue their activity and present for evaluation after competition or training. Based on military studies, knee immobilization with the knee flexed to 120 degrees for the initial 24 hours is recommended.¹⁸ Severe contusions may benefit from flexion immobilization for up to 48 hours.^{12,18} This can be achieved with either a hinged knee brace with locking capabilities, a compressive elastic wrap, or an anteriorly placed splint. Ice and compression should also be initiated in the first 24 to 48 hours, followed by elevation (the PRICE protocol). After the initial period of flexion, active, pain-free ROM exercises should be initiated. For severe contusions, crutches may be used until quadriceps control has returned. In these cases, physical therapy may be warranted, as well as the use of electrical stimulation. Ryan et al.¹² demonstrated a shortened recovery time for the moderate and severe groups from 56 to 19 days and from 72 to 21 days, respectively, via utilization of the aforementioned protocol. After return to pain-free motion, patients are allowed to return to noncontact sport-specific training. Return to contact sports is allowed after the thigh firmness has resolved and the muscle is not tender to palpation. If the patient is involved in a contact sport, a thigh pad with a ring should be recommended to minimize recurrence.^{12,18}

Nonsteroidal antiinflammatory drugs (NSAIDs) are used initially to decrease pain and swelling, but their long-term use is discouraged. However, for persons with severe contusions or contusions with firm masses that do not dissipate in the initial 7 to 10 days, NSAIDs may be used to prevent heterotopic bone formation or myositis ossificans. Although randomized controlled studies have not been performed to examine the effectiveness of NSAIDs in the treatment of myositis ossificans, their presumed effectiveness is based on animal studies and reports of their use in patients who have had a total hip replacement.¹⁸⁻²⁰

Operative treatment. Operative treatment of quadriceps contusions is primarily reserved for persons with compartment syndromes.²¹⁻²³ Aspiration or surgical decompression of a thigh hematoma has been reported; however, no literature is available to support the effectiveness of these treatments.²¹ Aspiration of knee effusions associated with severe contusions may decrease painful ROM, but support of this treatment is anecdotal. Other surgical considerations are reserved for partial or complete quadriceps tears or avulsion off the patella. Operative treatment of myositis ossificans should be reserved for patients with symptomatic loss of motion, pain, and strength once bone maturation has been demonstrated on a three-phase bone scan. Such treatment usually is performed 12 to 24 months after the time of injury.²⁴⁻²⁶

Postoperatively, compressive dressings, ice, rest, elevation, and standard surgical site care should be instituted. Again, early ROM

is critical to increase healing and decrease the risk of heterotopic bone formation.

Decision-Making Principles

The ROM classification system proposed by Jackson and Feagin¹¹ currently serves as the gold standard for the treatment and prognosis of thigh contusions. As a rule, nonoperative treatment should be pursued when possible provided compartment syndrome has not been identified. The aforementioned military studies have demonstrated the quickest return to activity, with immediate flexion of the affected knee to 120 degrees during the initial 24 to 48 hours from the time of injury. Subsequent passive and active ROM should follow to prevent heterotopic bone formation and increase rates of RTP.^{12,14} Heterotopic bone formation should be managed with the previously outlined contusion rehabilitation protocol and may be monitored with triple-phase bone scans to evaluate for skeletal maturity. Surgical intervention should be considered for myositis ossificans only if loss of strength and ROM has not resolved during the ensuing months. Bony maturation must be identified prior to excision to prevent local recurrence.²⁶⁻²⁸



Authors' Preferred Technique

Quadriceps Contusion

Timely evaluation of the patient should be undertaken to assess the severity of the contusion and rule out more severe sequelae, such as compartment syndrome. Ice, knee flexion, and a gentle compressive wrap are instituted if assessment takes place on the field. For patients with moderate to severe contusions, we prefer to institute immediate knee flexion to 120 degrees with a locked, hinged knee brace. Use of ice and a compressive wrap or sleeve is also instituted with rest. The patient is reevaluated within 24–48 h by the treating physician or a therapist, and active ROM of the knee is begun, with a greater than 120-degree return to ROM desired as soon as possible. Electrical stimulation is added as warranted. If no contraindication exists, NSAIDs are begun at prescription-strength dosage for 5–7 days around the clock. However, they are discontinued if the patient cannot tolerate them. Noncontact training is permitted once quadriceps control and painless full ROM has returned, which is typically seen in 10–14 days. Full contact is permitted at the 3- to 4-week mark provided no further setbacks occur and if a thigh pad is worn. For patients who demonstrate worsening symptoms at the 14- to 21-day mark, radiographs are repeated and return to sporting activities is tabled.

Return to Play

A mild contusion may be managed symptomatically with ROM exercises and NSAIDs, thus allowing the athlete to continue in competition or training. For moderate or severe injuries, non-contact training is permitted once quadriceps control and painless full ROM has returned, which is typically seen in 10 to 14 days. Full contact is permitted at the 3- to 4-week mark provided that no further setbacks occur and a thigh pad is worn. The patient should be seen and cleared by a physician, and dedicated strength testing may be warranted to ensure the patient's safety.¹¹⁻¹³

Results

With some exceptions, the literature on management and outcomes after thigh contusions is mostly based on case studies or

anecdotal evidence. Nonetheless, most reports note good to excellent results, with almost all athletes retuning to play. Ryan et al.¹² demonstrated a shortened recovery time for the moderate and severe groups from 56 to 19 days and from 72 to 21 days, respectively, with utilization of immediate flexion of the knee to 120 degrees, followed by early ROM exercises. These findings were duplicated by Aronen et al.¹⁸

Myositis Ossificans

Myositis ossificans is heterotopic ossification in an area of muscle, soft tissue, or disrupted periosteum (Fig. 87.4). It is generally associated with a severe contusion and has a reported occurrence rate of 9% to 14%.²⁹⁻³¹ In their military studies, Ryan et al.¹² identified the greatest risk factors for myositis ossificans: knee flexion less than 120 degrees, sustaining the injury while playing football, having had a previous quadriceps injury, experiencing a delay in treatment greater than 3 days, and having an ipsilateral knee effusion.

Although this disease process can occur without a history of trauma, athletes can usually describe a sentinel event that caused a hematoma formation. The hematoma organizes, and calcium deposits are formed by the body. In a process that is not entirely understood, osteoblasts invade the formed hematoma and begin to make bony spicules.³⁶ This process tends to occur near joints and at tendon origins, but it can occur anywhere along the course of a muscle. The process can start as soon as 1 week after injury and can be detected on plain films a minimum of 3 weeks after injury. Patients present with a rapid enlargement within the soft tissues, decreased ROM, and significant pain 1 to 2 weeks after injury. The patient has swelling and warmth at the site as well as an increased erythrocyte sedimentation rate and serum alkaline phosphatase level.²⁶ Any treatment modalities implemented should not promote hematoma formation. Massage and manipulation should be avoided. Therapy should consist of active stretching and strengthening. Passive ROM should be delayed for at least

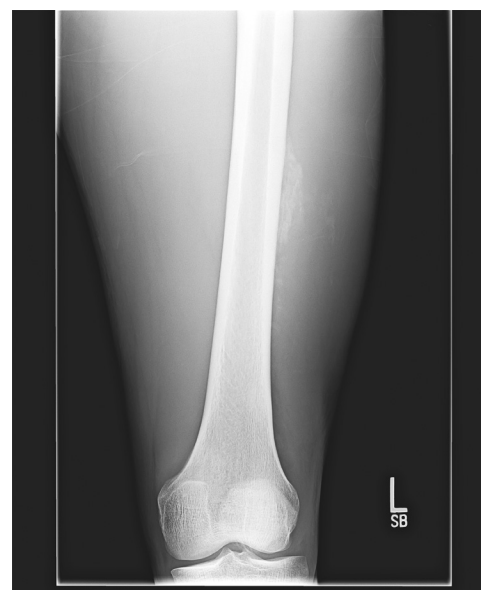


Fig. 87.4 Anteroposterior view of the femur demonstrating myositis ossificans.

3 to 6 months.³⁷ For patients with refractory loss of ROM, surgery should be considered. Surgery, if warranted, should be delayed for at least 9 to 12 months to allow the lesion to mature. A bone scan can help ascertain the maturity of the lesion. Patients and clinicians should be aware that the lesion may recur despite surgical removal.

Complications

Thigh compartment syndrome is a rare complication associated with quadriceps contusion, and treatment of this condition remains controversial. Most authors advocate emergent fasciotomies, but some reports in the literature advocate nonsurgical intervention, even when compartment pressure standards are met for a diagnosis of compartment syndrome.^{21-23,32} The rationale for this proposal is derived from Robinson et al.³² and others, who note that in cases of sports-related thigh compartment syndromes with pressures greater than 55 mm Hg, no adverse sequelae were identified at 1 year with a return to preinjury strength and ROM. Standard compartment syndrome measurement should be taken, and identification is made on the basis of one of the following criteria as identified in various studies: compartment pressures of 30 mm Hg, 45 mm Hg, or less than 30 mm Hg difference between compartment pressure and diastolic blood pressure.³³⁻³⁵

Future Considerations

Future consideration should be given to the pursuit of level I and II studies on the management of quadriceps contusions. Additionally, exploration of the nonoperative management of thigh compartment syndrome is also warranted.

Groin Contusions

Groin contusions involve the adductor musculature and usually occur from a direct blow to the inner thigh. They are often seen in soccer players and cyclists (Fig. 87.5). Aside from the standard contusion treatment of ice, NSAIDs, physical therapy, and gradual RTP, the treating clinician should be aware of the possibility that vascular complications may develop, such as phlebitis and thrombosis. Ultrasound is a useful noninvasive method of diagnosing vascular complications.

STRAINS

Muscle strains are among the most common athletic injuries, representing 30% to 50% of all injuries.^{26,38,39} Most strains occur at the myotendinous junction in the fast-twitch type 2 muscle fibers of biarticular muscles undergoing eccentric contraction. Although most strains occur at this interface, muscle strains can

occur anywhere along the length of the muscle.^{40,41} Over a 10-year collection period in the NFL, Feeley and colleagues⁴² noted that 41% of strains involved the hamstrings, 28% involved the quadriceps, and 17% involved the groin.

Classification

The clinical classification system describes the severity of the muscle injury: mild, moderate, or severe (Table 87.2). Mild (grade I) strains involve tearing of a few muscle fibers with mild pain and minimal loss of strength. Moderate (grade II) strains involve increased tearing of muscle fibers with some loss of strength. Severe (grade III) strains include tearing of the entire muscle with complete loss of strength.^{43,44} Occasionally in adolescents and a small subset of adults, the tendons of origin or insertion may be avulsed; these are classified as grade IIIB.

Quadriceps Strains

The only biarticular muscle in the quadriceps is the rectus femoris, but all muscles receive innervation from the femoral nerve. The direct head of the rectus femoris gives rise to a central tendon in the proximal thigh, allowing strains to occur proximal to the musculotendinous junction. The primary function of these muscles is knee extension.¹⁸

Quadriceps strains typically affect the rectus femoris. These strains can occur proximally or distally. The quadriceps is commonly affected for several reasons: it crosses two joints, has a high percentage of type II fibers, and has a complex musculotendinous architecture. Sudden forceful eccentric muscle contraction of the quadriceps is required during hip extension and

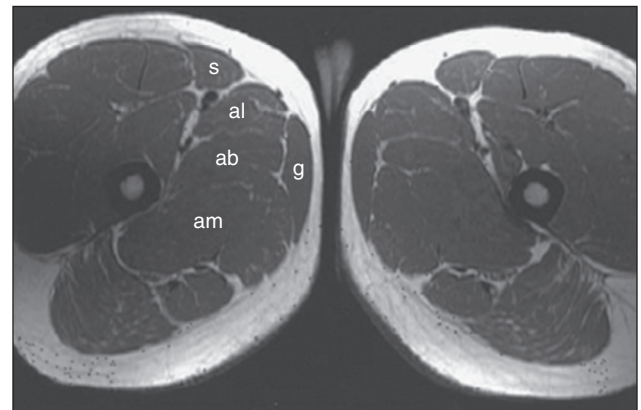


Fig. 87.5 Cross-sectional anatomy of the thigh on a T1-weighted axial magnetic resonance image. The musculature of the medial thigh includes the adductor longus (*al*); adductor brevis (*ab*); adductor magnus (*am*); gracilis (*g*); and sartorius (*s*).

TABLE 87.2 Clinical Grading System for Strains

Grade	Muscle Tearing	Strength Loss	Pain	Physical Examination: Muscle Defect
I	Mild	Minimal/none	Mild	Absent
II	Moderate	Moderate/severe	Moderate/severe	Possible palpable defect
IIIA	Severe	Severe	Severe	Frequent palpable defect (complete rupture of musculotendinous unit)
IIIB	Severe	Severe	Severe	Frequent palpable defect (avulsion fracture at the tendon's origin or insertion)

knee flexion and can lead to increased forces across the muscle-tendon interface, resulting in strain.^{38,45-47}

History

Athletes with a quadriceps strain typically play sports that require cutting, jumping, or kicking, such as soccer, rugby, football, and basketball. A thorough history will tend to reveal the onset of anterior thigh pain either during the maximal extension phase of the thigh, after a sudden change in direction or during deceleration after a forceful kick.⁴⁸ Often patients will have an antalgic gait, loss of knee flexion, or an anterior thigh mass.^{45,49-52} Most commonly quadriceps strains occur distally at the musculotendinous junction; however, they can also occur proximally or centrally.⁵¹ Pain may be reported immediately after the injury has occurred, but often the athlete may be able to continue playing through a practice or game with the onset of symptoms occurring after a cooling-down period.^{43,46,53}

Physical Examination

Initially physical examination is best carried out with the patient lying supine; it concludes with a prone examination. With the patient supine, a thorough palpation of the muscle should be undertaken to assess for tenderness, swelling, masses, or defects. Ecchymosis may occur 24 hours or more after the onset of injury, after which ROM of the knee and hip should be performed and the injured knee compared with the contralateral side. Next, evaluation of knee extension should be undertaken with the hip flexed to 90 degrees and then with the hip in extension. Weakness identified in hip flexion indicates injury to the rectus, whereas weakness in hip extension points toward a vastus injury. It should be noted that rectus injuries have a slower recovery time than do those that occur in the vastus.^{24,54} Once the supine examination is complete, the patient is positioned prone and strength is retested, which can help to isolate the quadriceps for motion and strength evaluation. For moderate to severe strains, pain is usually felt with resisted knee extension.

Imaging

Imaging in quadriceps strains is typically used more as an adjunct than for diagnosis. Radiographs, ultrasound, and MRI may all be used and can provide information regarding the length of recovery. Radiographs are routinely normal; however, in the young athlete they can be particularly important in identifying avulsion fractures or stress fractures. Ultrasound and MRI are most beneficial in predicting recovery time. Finlay and Friedman⁷⁰ reported that ultrasound is a highly sensitive and specific method for evaluating acute quadriceps injuries. The use of dynamic ultrasound with the knee in flexion and extension can allow differentiation between hematoma and muscle tears; however, ultrasound is highly operator-dependent. MRI is still considered to be the gold standard for quadriceps evaluation (Fig. 87.6),⁵⁵⁻⁵⁷ although its use in the initial evaluation phase is usually restricted to high-level or professional athletes to better predict a RTP. For recreational athletes, MRI is indicated if the symptoms do not improve after 2 to 3 weeks of rehabilitation or for evaluation of chronic (>8 weeks) strains.⁴⁰ Cross et al.⁵³ reported that MRI can estimate the size of the quadriceps strain

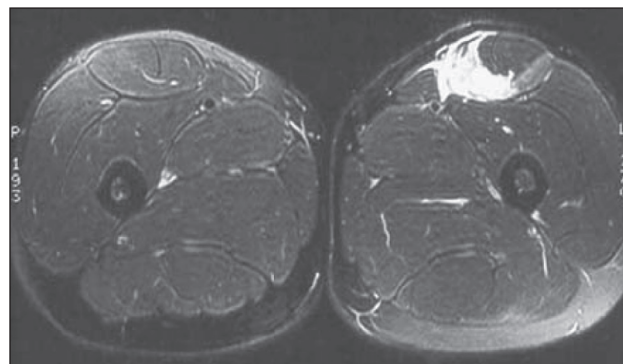


Fig. 87.6 A T2-weighted axial magnetic resonance image of the thighs showing increased fluid in the rectus femoris, indicative of a quadriceps strain.

TABLE 87.3 Treatment Protocol for a Quadriceps Strain

Phase	Activity
I	RICE, NSAIDs
II	Isometric exercises
III	Isotonic exercises
IV	Isokinetic exercises
V	Running, plyometrics, jumping exercises
VI	Sport-specific training

NSAIDs, Nonsteroidal antiinflammatory drugs; RICE, rest, ice, compression, and elevation.

and predict the duration of rehabilitation. Furthermore, involvement of the central tendon signifies a significantly longer rehabilitation interval.⁵³

Treatment Options

Most quadriceps strains can be managed nonoperatively (Table 87.3). In the acute phase, the treatment of quadriceps strains begins with the standard PRICE protocol in an effort to decrease swelling and hematoma formation. Severe strains may require a short period of crutch use. Passive, pain-free stretching should also be initiated in the first 24 to 48 hours.^{24,54} At the 3- to 5-day mark, the active rehabilitation phase may begin. First, isometric exercises and increasing movement within a pain-free arc of motion are begun. Eccentric contraction of the quadriceps should be avoided. As strength and ROM continue to improve, the patient is transitioned to isotonic training with increasing resistance as tolerated.²⁴ Once strength has improved, isokinetic exercises should focus on low resistance and high speeds. A progressive running and kicking program should then be initiated, starting with a slow jog and advancing to sprinting. When sprinting is tolerated, kicking may be started with a light ball over short distances, advancing to a normal-weight ball over longer distances.⁵³ Finally, sport-specific training should begin. The patient may RTP after completion of this training but should continue to work on a quadriceps rehabilitation program involving stretching, isokinetic strengthening, and conditioning.

Surgical intervention is reserved for total or near-total muscle tears, persistent pain with nonoperative treatment, reattachment



Fig. 87.7 Quadriceps rupture with retraction.

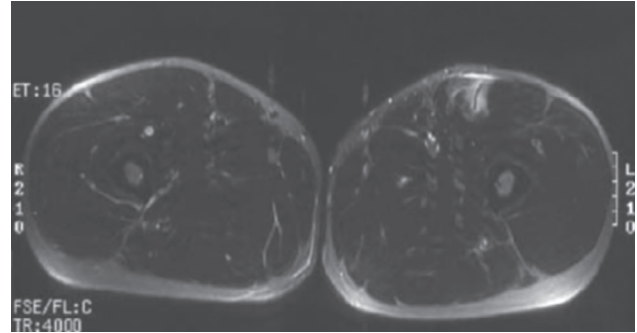


Fig. 87.8 An acute bull's-eye lesion.

of avulsion fractures, and large hematoma evacuation. Time until RTP after rectus reattachment surgery was noted to be 9 months in one report (Fig. 87.7).^{24,48,58}

Decision-Making Principles

Nonoperative treatment should be pursued whenever possible for quadriceps strains. There are very few indications for surgical intervention. Moreover, controversy exists regarding the value of surgical decompression of large hematomas as well as the need for operative intervention for rectus avulsion injuries. Complete muscle tears, true fractures associated with quadriceps strains, or symptomatic mature myositis ossificans appear to be the only criteria for surgical intervention. However, based on some published reports, recalcitrant nonoperative treatment of severe quadriceps strains may also warrant surgical intervention.⁴⁸



Authors' Preferred Technique

Quadriceps Strains

We prefer to use nonsurgical management. In the initial 24–72 h, PRICE, NSAIDs, and gentle ROM are begun. If sufficient progress is noted, the rehabilitation protocol is then begun at day 3–5, with a focus on pain-free motion. If at any point a setback is experienced, the protocol is backed down. Antiinflammatory modalities are also instituted by the trainer or physical therapist, along with ultrasound and electrical stimulation for moderate to severe strains. Operative intervention is considered only for avulsion fractures that are symptomatic for 12 months or more. Gentle passive ROM is begun 7–10 days after the surgery, with an emphasis on avoiding eccentric muscle contractions. Once patients can tolerate this gentle passive ROM, they may begin the aforementioned rehabilitation protocol.

Return to Play

A review of the literature by Orchard et al.⁵⁴ notes a lack of consensus for safe RTP after muscle strains. Despite this finding, certain criteria that appear in the literature seem to lower the risk of reinjury after a quadriceps strain. ROM of the knee should be equal to that of the contralateral leg. Isokinetic quadriceps

strength should fall within 15% of the contralateral side. Ideally, a sport-specific program should be completed, including observed straight-ahead sprinting and figure-eight cutting sprints. Most patients are able to RTP within 2 to 3 weeks while wearing a compressive thigh wrap and continuing a dedicated stretching and strengthening program.

Results

Good to excellent results have been reported with use of the phased treatment protocol for quadriceps strain. Recent articles suggest that higher-level athletes should resume play earlier with the understanding that the risk of reinjury is then higher. Stratification of the risk of reinjury can be aided by imaging, such as ultrasound or MRI, to quantify the severity of the initial injury.⁵⁴ Based on imaging, injuries involving greater than 15% of the rectus cross-sectional area have demonstrated a longer recovery phase (14.6 days vs. 8.9 days). Strains greater than 13 cm in length also resulted in a doubling of recovery time. Contrast enhancement around the central tendon, the so-called acute bulls-eye lesion, also revealed a significantly worse prognosis (26.8 days vs. 9.2 days) (Fig. 87.8).^{41,45} Although many investigators have recommended surgical treatment for proximal avulsion fractures, Hsu and coworkers⁵⁸ treated two NFL kickers with a nonoperative regimen and noted that both were able to return to full practice after 3 to 12 weeks.

Limited data exist for surgical outcomes of quadriceps strains, with the majority of the information being case reports. Straw et al.⁴⁸ reported a full return of strength and return to competition after repairing a chronic proximal rectus avulsion in a semiprofessional soccer player. Similar reports exist in the literature, noting a return to previous activity level with nearly symmetric strength in athletes treated surgically for chronic rectus weakness.⁵⁸

Complications

Complications associated with quadriceps strains include myositis ossificans, reinjury, residual weakness, and compartment syndrome. Risk stratification for reinjury is used only as a guide, further emphasizing the need for dedicated RTP criteria and a thorough warmup and stretching program. Residual weakness that persists for more than 12 months should warrant evaluation of surgical intervention. Myositis ossificans has previously been discussed in the contusion section, and urgent intervention in the initial 24 to 72 hours for severe strains is warranted.

Future Considerations

The future of quadriceps strains lies in identifying more rapid improved RTP criteria while limiting reinjury risk. Some have advocated removing the athlete from competition completely until all symptoms have resolved. However, others are advocating a RTP even before the full RTP criteria have been met. Anecdotally it is the physician's charge to protect the athlete and limit short- and long-term risk.

Adductor Strains

Adductor strains, which are the most common groin injuries in athletes, occur as a result of a forced push-off or side-to-side motion.^{59,60} These injuries are seen more often in athletes who play soccer, hockey, rugby, martial arts, and football. Their reported incidence ranges from 13% to 43%.⁶¹⁻⁶⁴ Adductor strains are commonly the result of a forceful abduction of the thigh during an intentional adduction movement. The musculotendinous junction of the adductor longus is most commonly involved.^{60,65} Inflexibility, previous injury, and strength imbalance between adductors and abductors have been implicated as risk factors for adductor strains.

Adductor strains can be graded from I to III. Mild (grade I) strains result in pain with minimal strength loss, moderate (grade II) strains result in strength loss and pain, and severe (grade III) strains result in complete functional and motor loss.

History

The athlete will typically present with a chief complaint of acute groin pain, tenderness, and swelling. The mechanism may be a sudden change in direction, or the athlete may note that he or she was attempting a forceful kick that was met with opposite resistance, such as an opposing player blocking the kick. Pain will be noted to be worse with side-to-side movement. Often the patient has a history of an adductor injury. A review of the literature suggests the need for a thorough history and examination because groin pain in 27% to 90% of athletes has more than one cause.^{66,67} A differential of groin pain includes osteitis pubis, athletic pubalgia, hernias, femoroacetabular impingement, and stress fractures.

Physical Examination

Patients with moderate to severe injuries may display an antalgic gait; in rare instances they may not be able to bear weight on the affected limb. A complete examination with palpation of the medial groin should be performed to look for tenderness, swelling, ecchymosis, and defects. ROM and strength should also be assessed. Medial groin pain is accentuated with resisted hip adduction. In addition, the pubic rami and inferior abdominal wall should be inspected to rule out other causes of pain.

Adductor strains must be differentiated from other sources of groin pain. Tenderness to palpation at the pubis may result from osteitis pubis, athletic pubalgia, or avulsion injury of muscular origin.⁶⁸ Osteitis pubis results in tenderness over the pubic tubercle or inguinal ring and often can be differentiated because symptoms exacerbate with situps.^{66,69} Persons with athletic pubalgia may have tenderness over the pubic tubercle, and this tenderness can

be reproduced with resisted situps or the Valsalva maneuver. An inguinal hernia examination should also be performed. Pain with provocative hip testing—as with the flexion, adduction, internal rotation (FADIR) maneuver—may indicate femoroacetabular impingement or hip labral tearing.

Imaging

Plain radiography with the patient standing on one leg is the initial imaging modality of choice. These radiographs are acquired less for diagnosis of adductor strains and more to rule out other pathology, such as avulsion fractures or osteitis pubis. A bone scan, MRI, and ultrasound have also been used to varying degrees (Fig. 87.9). An MRI is not routinely ordered and is used only as a guide in estimating the length of recovery. Much as with other strains of the thigh, involvement of greater than 50% of the cross-sectional area, tissue fluid collection, and deep muscle tears are predictive of a slower return to sport.³⁸ Ultrasound has been used to evaluate the musculotendinous junction and to aid in the differentiation of a strain from other conditions, including hernia.⁷⁰

Treatment Options

Nonoperative. Adductor strains are managed similarly to quadriceps and hamstring strains. The majority of these injuries are treated nonsurgically. PRICE, NSAIDs, and a progressive rehabilitation program are initiated. Typically persons with grade I strains do not seek medical attention; however, if they do seek treatment, a 3- to 7-day course of progressive ROM and strengthening is recommended. Sport-specific drills should be performed and observed prior to clearance. For patients with more severe strains, early stretching sometimes has been noted to aggravate the injury. A gentle progression through isometric, isotonic, and

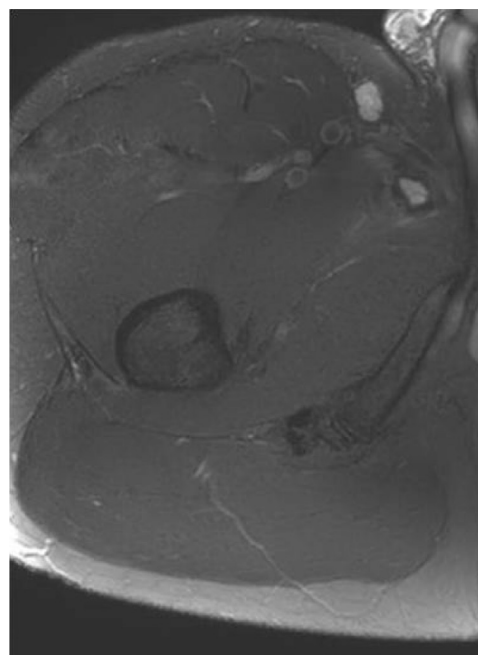


Fig. 87.9 Cross-sectional T2 magnetic resonance image of the pelvis showing a complete tear of the proximal adductor longus.

isokinetic exercises should be initiated, and the patient should be noted to be pain-free before advancing to sport-specific drills. Upper-body and core exercises may begin in phase II. Swimming and use of a stationary bike and elliptical machine may begin in phase III. Sports-specific drills may begin once the patient has ROM and adductor strength that is 75% of ipsilateral abductors. Emphasis should be placed on reinjury prevention via a focused stretching and strengthening program designed to create an adduction-to-abduction strength ratio of 0.8 or higher.⁷¹ Corticosteroid injections and platelet-rich plasma (PRP) injections have yet to yield sufficient data to be considered first-line treatment options.

Operative. Surgical intervention is rarely required for adductor strains. Strains recalcitrant to conservative care with persistent pain and weakness may require tenotomy versus repair. The literature for surgical intervention is predominately based on case reports. Patients with complete avulsions who have had persistent symptoms for 6 months or longer may be treated with repair. For incomplete tears that are symptomatic, tenotomy may be best indicated.^{60,72}

Decision-Making Principles

Nonoperative treatment is the overwhelming option of choice for adductor strains. Although a high reinjury rate applies, surgical intervention is very rarely warranted and has had mixed results. Proper education of the athlete and the trainer as to the rehabilitation course and future prevention is the key to a successful outcome. Surgical intervention is considered only for complete avulsions that hinder RTP for 6 months or more or for chronic partial tears that may require tenotomy after a similar length of time.



Authors' Preferred Technique

Adductor Strains

We prefer nonsurgical management, with an emphasis on normalizing abductor-adductor strength imbalance while rehabilitating the athlete. The athlete is initially treated with the standard PRICE protocol and a 5- to 7-day course of NSAIDs. Beginning on day 3, passive ROM and gentle isometric exercises are begun. Eccentric loading is avoided in phase I. Advancement to phase II is permitted once the athlete can adduct against gravity without pain. Phase II consists of isotonic exercises under supervision. Stationary biking and swimming are permitted, as is core and upper-body training. Stretching, which is typically avoided in phase I, is permitted to the level of discomfort without pain. Attention to abductor-adductor strengthening is begun. Progression to phase III occurs once the patient demonstrates pain-free strength against submaximal abduction-adduction resistance. In phase III, isokinetic training is incorporated, followed by sport-specific drills. A dedicated warmup and stretching program should be instituted to focus on reinjury awareness for the athlete. RTP is permitted once single-leg strength is demonstrated to be equal to the contralateral side in all planes and the adduction-to-abduction strength ratio is greater than 80%.

Return to Play

Return to competition is permitted once pain-free sport-specific movements can be performed with adductor strength within 20% of the contralateral side. Although most patients

RTP in 2 to 3 weeks, severe cases may take 8 to 12 weeks. Dedicated warm-up stretching should be emphasized, as should in-season strengthening. RTP prior to pain-free sport-specific training has been associated with chronic adductor strains.⁷¹ Reinjury education and prevention is warranted as well. RTP after tenotomy for chronic adductor pain ranges from 8 to 12 weeks.

Results

Adductor strains treated with a dedicated rehabilitation program and injury prevention awareness have demonstrated excellent results. Tyler et al.⁷⁵ reported a greater than fourfold reduction in adductor strains in the National Hockey League after institution of a dedicated adductor strengthening and stretching program. Holmich et al.⁷⁶ noted that active therapy in persons with chronic adductor strains provided better results than those seen in patients treated with passive therapy programs. RTP rates after surgical tenotomy for chronic adductor strains have been demonstrated to range from 60% to 90%. Concomitant injuries, such as sports hernias and subsequent repair, have been shown to affect outcome.^{73,74}

Complications

The most common complication of adductor strains is reinjury. Seward et al.⁷⁷ and Tyler and associates⁷⁵ report recurrence rates of 32% and 44%, respectively. The highest risk factor for adductor strain is an adductor-to-abductor strength of 80% or less.⁷¹ Other complications include myositis ossificans, compartment syndrome, calcific tendinitis, and chronic groin pain.

Future Considerations

Currently there is significant interest in the treatment of muscle, tendon, and ligament injuries with PRP. This involves using a commercially available kit to centrifuge a blood sample in order to concentrate platelets and reduce red blood cells.⁷⁸ Platelets, the first cell type to arrive at the site of injury, possess high quantities of growth factors, which they release to stimulate tissue repair. PRP therefore is replete with growth factors and cell signaling molecules that may be key to the regulation of tissue healing.⁷⁹ Multiple small studies exist that attest to its safety, but they demonstrate inconsistent results. PRP has shown promise in treating hamstring injuries among NFL athletes within 24 to 48 hours of acute injury. Mejia and Bradley⁸⁰ have shown faster RTP in grade 1 and 2 injuries without recurrence, which resulted in a one-game difference in favor of PRP. On the other hand, deVos et al.,⁸¹ in a randomized controlled trial, were unable to demonstrate improved pain or function in Achilles tendinopathy as compared with saline injection. These inconsistencies are confounded by variations in platelet and white blood cell (WBC) concentrations among the different kits, making it difficult to draw conclusions across studies.⁷⁸ Ultimately we are unable to recommend for or against the use of PRP in the treatment of hip and thigh muscle contusion or strain at this time but advocate for future level 1 studies, which may help to delineate its best use.

SPECIAL CONSIDERATIONS FOR THE PEDIATRIC AND ADOLESCENT ATHLETE

Avulsion injuries are uncommon in the pediatric and adolescent populations and are easily misdiagnosed as muscle strains or apophysitis without a high degree of clinical suspicion and vigilance. They occur during a forceful muscle contraction and result from failure through a secondary apophysis rather than a typical strain, which usually occurs at the myotendinous junction.^{82,83} Most of these apophyseal ossification centers are closed by age 17 except for the Anterior superior iliac spine (ASIS), which can remain unfused until age 25.⁸⁴ The incidence reflects a gender bias, with males accounting for 66% to 86% of injuries. The most common sites affected are the anterior inferior iliac spine (AIIS) at 33%, ischial tuberosity at 31%, and ASIS at 25%.⁸⁵

History

Athletes with avulsion fractures will typically be participating in a sport that requires sprinting, jumping, or kicking, such as football, basketball, or soccer.^{82,83,86} Most are able to describe a distinct moment of injury accompanied by a pop or crack, followed by pain. This can be differentiated from apophysitis, which is a chronic overuse injury and presents with a more gradual onset.⁸⁷

Physical Examination

The physical exam can be conducted in much the same way as with muscle strain. Observation of gait, palpation of the affected area, and neuromuscular exam are all key components of the clinical visit. Swelling and tenderness to palpation are typical of these injuries, as is muscle weakness and pain with passive stretch.

Imaging

Although it is understandable to want to limit radiation in this population, plain radiographs have been shown to diagnose avulsion fractures in 99% of cases and should be used, in preference to MRI or CT, if there is clinical suspicion.⁸⁸

Treatment

The majority of these patients can be treated nonoperatively with a dedicated rehab protocol. Schuett et al.⁸⁸ showed that 86% of patients had complete resolution of pain at 3 months. A 5-stage protocol has been demonstrated to be consistently effective in the majority of these patients. This protocol begins with 1 week of non-weight bearing, ice, and NSAIDs, then 2 to 3 weeks of passive ROM and partial weight bearing. Full weight bearing and isokinetic strengthening are then initiated at 4 weeks after injury. Following this, some authors allow sport-specific activities, whereas others begin resistance training. A return to sport is allowed at 2 months after injury.^{83,89} Operative treatment is typically reserved for failed nonoperative management. This includes persistent pain, symptomatic nonunion, or hip impingement after AIIS avulsion.

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For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

Ryan JB, Hopkinson WJ, Wheeler JH, et al. Quadriceps contusions. West Point update. *Am J Sports Med*. 1991;19(3):299–304.

Level of Evidence:

II

Summary:

The authors of this article provide defining data for the appropriate management of quadriceps contusions, altering previous standards of care. Current treatment regimens are predicated on the findings presented in this article.

Citation:

Beiner JM, Jokl P. Muscle contusion injury and myositis ossificans traumatica. *Clin Orthop Relat Res*. 2002;403S:S110–S119.

Level of Evidence:

II

Summary:

The authors provide an analysis of the etiology, treatment, and outcomes of myositis ossificans.

Citation:

Orchard J, Best TM, Verrall GM. Return to play following muscle strains. *Clin J Sport Med*. 2005;15(6):436–441.

Level of Evidence:

III

Summary:

Consensus regarding return-to-play guidelines has been lacking in the literature. This article provides treatment parameters and offers a perspective on earlier return to competition.

Citation:

Beiner JM, Jokl P. Muscle contusion injury and myositis ossificans traumatica. *Clin Orthop Relat Res*. 2002;403S:S110–S119.

Level of Evidence:

III

Summary:

The authors provide a comprehensive review of hamstring injuries, treatments, and outcomes. Gaps within the literature on hamstring injury are also identified.

Citation:

Lovell G. The diagnosis of chronic groin pain in athletes: a review of 189 cases. *Aust J Sci Med Sport*. 1995;27(3):76–79.

Level of Evidence:

III

Summary:

The author provides a systematic review evaluating the effectiveness of exercise therapy for groin pain in athletes and also considers the treatment of this disorder.

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